Kanageshwaran Dhakshinamoorthy

Stat 50

① HW #11
Section 4.4

①

$x \sim H(25, 10, 9)$

$$P(X=3) = \frac{\binom{10}{3}\binom{25-10}{9-3}}{\binom{25}{9}}$$

$$= 0.29398 = 0.2940$$

②

(a) $x \sim H(30, 4, 10)$

$$P(x=2) = \frac{\binom{4}{2}\binom{30-4}{10-2}}{\binom{30}{10}}$$

$$= 0.31199 = 0.3120$$

(b)

$$P(x=0) = \frac{\binom{4}{0}\binom{30-4}{10}}{\binom{30}{10}}$$

$$= 0.17679 = 0.1768$$

(3) $x \sim Geom(0.1)$

$P(x=12) = (0.1)(1-0.1)^{12-1}$

$= 0.03138 = 0.0314$

(4)

(a) $x \sim Geom(.40)$

$P(x=3) = (.40)(1-.40)^{3-1}$

$= 0.144$

(b) $P(x \leq 3) = P(x=3) + P(x=2) + P(x=1)$

$= (.4)(1-.4)^{3-1} + (.4)(1-.4)^{2-1} + (.4)(1-.4)^{1-1}$

$= 0.784$

(c) $\mu_x = 1/p$

$\mu_x = 1/.4 = 2.5$

(d) $\sigma_x^2 = \frac{1-p}{p^2} = \frac{1-0.40}{(0.40)^2}$

$\boxed{\sigma_x^2 = 3.75}$

(5) $Y \sim NB(r,p) \Rightarrow Y \sim NB(3, .4)$

(a) $P(Y=7) = \binom{7-1}{3-1} (.4)^3 (1-0.4)^{7-3}$

$= 0.124416 = 0.1244$

(b) $\mu_y = r/p$

$= 3/.4 = 7.5$

(c) $\sigma_y^2 = \frac{r(1-p)}{p^2} = \frac{3(1-.40)}{(.4)^2}$

$= 11.25$

(6) $x \sim MN(10, .50, .10, .40)$

$P(x_1 = 4, x_2 = 1, x_3 = 5)$

$= \frac{10!}{(4!)(1!)(5!)} (.5)^4 (.1)^1 (.4)^5$

$= 0.08064$

(9) $x \sim Geom(.8)$

(a) $P(x \leq 3) = P(x=3) + P(x=2) + P(x=1)$

$= (.80)(1-.80)^2 + (.80)(1-0.80)^1 + (.80)(1-.08)^0$

$= 0.992$

(b) P(x when 2 hours undetected and chances of getting detected in next hour)

⇒ The probabilities of each trial are independent.

$\therefore \boxed{P = 0.8}$

(c) $\mu_x = 1/p$

$= 1/.8 = 1.25$

$\boxed{\mu_x = 1.25}$

(10)

(a) $\mu_x = r/p$, where $X \sim NB(5, 1/1000)$

$\mu_x = \frac{5}{(1/1000)} = 5000$

$\boxed{\mu_x = 5000}$

(b) $\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\frac{r(1-p)}{p^2}} = \sqrt{\frac{5(1-\frac{1}{1000})}{(1/1000)^2}}$

$= 2234.9\cancel{8}49664$

$= \cancel{8}2234.95$

11

(a) $P(x=3)$, $X \sim H(N,R,n) \Rightarrow X \sim H(15,12,4)$

$$P(x=3) = \frac{\binom{12}{3}\binom{15-12}{4-3}}{\binom{15}{4}}$$

$$= 0.483516 = 0.4835$$

(b) $$\mu_x = \frac{nR}{N}$$

$$\mu_x = \frac{4(12)}{15}$$

$$\boxed{\mu_x = 3.2}$$

(c) $$\sigma_X = \sqrt{\sigma_X^2} = \sqrt{n\left(\frac{R}{N}\right)\left(1-\frac{R}{N}\right)\left(\frac{N-n}{N-1}\right)}$$

$$= \sqrt{4\left(\frac{12}{15}\right)\left(1-\frac{12}{15}\right)\left(\frac{15-4}{15-1}\right)}$$

$$= \sqrt{0.502857143} = 0.7091$$

13

(a) $X \sim H(60, 5, 10)$

$$P(X>1) = 1 - P(X \leq 1)$$

$$= 1 - \left[\binom{5}{0}\binom{55}{10} \Big/ \binom{60}{10}\right] - \left[\frac{\binom{5}{1}\binom{55}{9}}{\binom{60}{10}}\right]$$

(6)

$= 0.1904$

(b) $X \sim H(60, 10, 10)$

$$1 - P(X \leq 1) = 1 - \left[\frac{\binom{10}{0}\binom{50}{10}}{\binom{60}{10}}\right] - \left[\frac{\binom{10}{1}\binom{50}{9}}{\binom{60}{10}}\right]$$

$= 0.5314$

(c) $X \sim H(60, 20, 10)$

$$1 - P(X \leq 1) = 1 - \left[\frac{\binom{20}{0}\binom{40}{10}}{\binom{60}{10}}\right] - \left[\frac{\binom{20}{1}\binom{40}{9}}{\binom{60}{10}}\right]$$

$= 0.9162$

(15)

(a) $X \sim (20, .25, .35, .40)$

$P(X_1 = 5, X_2 = 7, X_3 = 8)$

$$= \frac{20!}{5!\,7!\,8!}(.25)^5(.35)^7(.40)^8$$

$= 0.0408 = 0.0411$

(b) $X \sim Bin(20, .40)$

$$P(X \geq 10) = P(X=11) + P(X=12) + P(X=13) + P(X=14) + P(X=15) + P(X=16) + P(X=17) + P(X=18) + P(X=19) + P(X=20)$$

$$= 0.070994879 + 0.0355 + 0.0146 + 0.0049 + 0.0013 + 0.0003 + 0.0000 + 0.0000 + 0.0000 + 0.0000$$

$$= 0.1275.$$